Unit 4 – Behind Movements

Content :
* Different types of movements.
* Different types of muscles.
* Skeletal system.
* Structural frameworks in animals.
* Types of joints.
* Disorders of bones and muscles.
* Movements in plants.

Unit summary :

All living things have movement. But in living things, including microorganisms, diversity is shown in movements and the means it occur. Energy is required for movement and locomotion. In humans, muscles and bones help in movement and locomotion.

Muscle cells with the proteins actin and myosin have the ability to contract and relax. On the basis of muscle cells, human muscles are classified as skeletal muscle, smooth muscle, and cardiac muscle. These include both voluntary muscles and involuntary muscles.

The 206-bone human skeleton can be divided into axial and appendicular skeletons. A bone with a covering (periostium) contains blood vessels, nerves, lymph vessels, and osteoblast cells. Cartilage is a connective tissue that has no blood vessels or nerves and is softer and more flexible than the bone.

Humans mainly have an endoskeleton. Some animals have exoskeleton. A few animals without bones or cartilage have a hydroskeleton filled with water. Based on evolutionary history, it can be understood that after the emergance of organisms with hydroskeleton, organisms with exoskeletons and then endoskeletons were evolved.

Joints are the parts where bones are connected. Joints help in movement and locomotion. There are of four types of joints namely pivot joint, ball & socket joint, hinge joint and gliding joint. Each joint consists of a ligament, capsule, synovial membrane, synovial fluid and cartilage.

Nutritional factors and exercise are necessary for the growth and development of muscles and bones. Osteoporosis, rheumatoid arthritis, sprains, muscular dystrophy etc. are examples of skeletal muscular disorders.

Plants also move in response to stimuli. Those that follow the direction of stimulus are called tropic movements (tropism) and those that are not are called nastic movements. Apart from the phototropic, geotropic and hydrotropic movements, haptotropic and chemotropic movements are also take place in plants.

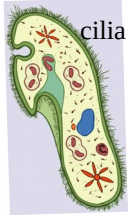
Questions & Answers :

1. Animals require more energy than plants. Reason?
   Unlike plants, animals use a good portion of their energy for movement.
2. Give examples of different types of movements seen in living things.

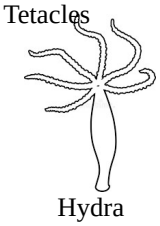
In microorganisms	Bacterium moves using its flagella, amoeba moves using pseudopodia, and paramoecium moves using its cilia.
In animals	Walking, running, peristalsis, heart beat, blood circulation, sperm movement
In plants	Seed germination, growth of plant parts, changes according to stimuli...

3. Examples of movements common to all living things?
   Movement of particles by osmosis, diffusion, active transport etc,
   Transport of gases, transport of nutrients, excretion of waste…
4. Means of movement in various organisms:

Bacteria - Flagellum Amoeba - Pseudopodia Paramecium - Cilia Hydra - Tentacle	Fish - Fin and tail Frog - Limbs Bird - Wing Whale - Fin and tail	Earthworm - Setae Cockroach -Wing, limb Lizard - Limbs

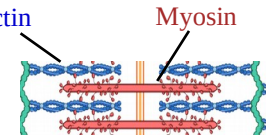


Paramoecium

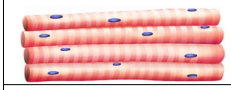
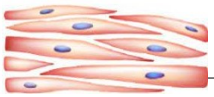
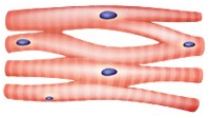


Hydra

5. Movement and means of movements in human beings ?
   - Cytoplasmic streaming (to distribute substances throughout the cytoplasm)
   - Pseudopodial movement (in WBCs, pseudopodia help in locomotion and defence)
   - Flagellar movement (the flagellum of the sperm helps to reach the egg)
   - Ciliary movement (the ovum moves by the movement of the cilia in the oviduct)
   - Muscular movement
6. What enables movement in our body?
   Bones and muscles.
7. What characteristic of muscles facilitates movement?
   Muscle cells can contract and relax.
8. Proteins that enable muscle cells to contract and relax?
   Actine, Myosin.

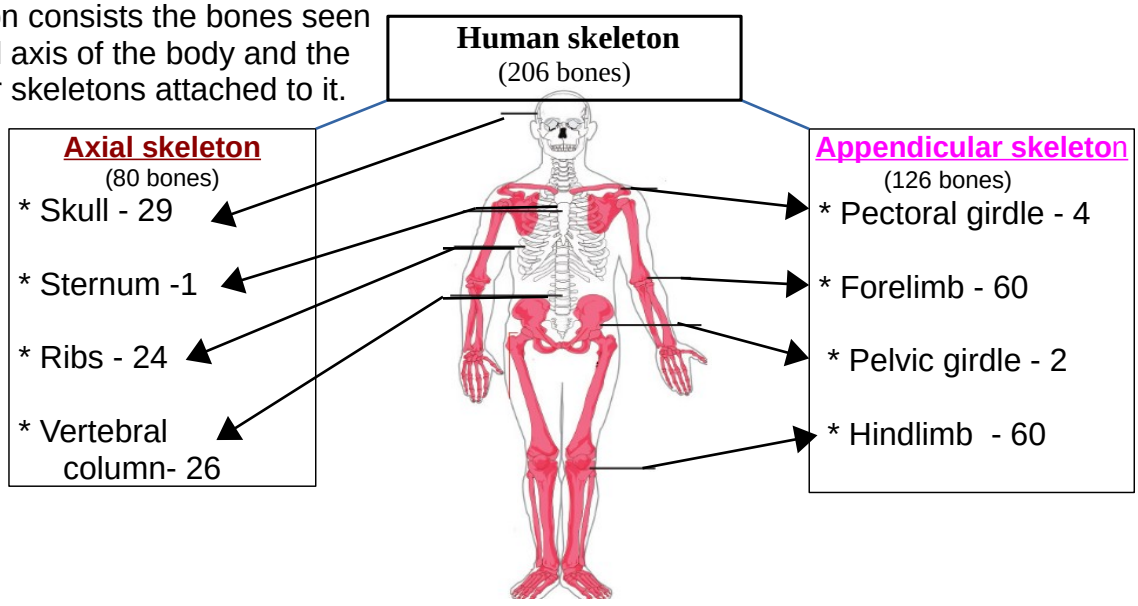


9. Different types of muscles in human body? What are the characteristics of each?

Skeletal muscle 	Attached to the bones. Cylindrical cells with striations, No branches. Voluntarily controlled muscles. eg:- Muscles in hands and legss, neck muscles, facial muscles..
Smooth muscle 	Seen in the hollow internal organs. Spindle shaped, single nucleated cells with out striations, No branches. Involuntary muscles. eg:- Muscles in intestine, stomach, blood vessels ..
Cardiac muscle	Muscles in the walls of heart. Cylindrical branched cells with striations, Inoluntary muscles. eg:- Muscles in heart walls. 

10. When skeletal muscles function continuously, muscle fatigue may happen. Give reason?
    When muscles work continuously, oxygen may completely used up and muscle become weak and lose the capacity to contract due to lactic acid formation by anaerobic respiration.
11. How muscles help us bend and straighten our limbs?
    The antagonistic actions of both the flexor muscle and extensor muscle help us fold and strech our hands or legs. Tendons also takes part in this action.
12. The parts that connect the ends of the muscles to the bones are ……..... ?
    Tendons.
13. Functions of skeletal system?
    Provide support, protection and shape to the body and aid in movement and locomotion.
14. The human skeleton has a total of ------ number of small and large bones.
    206. (Axial skeleton = 80, Appendicular skeleton =126)
    (About *18% of human body weight is constituted by the bones*)

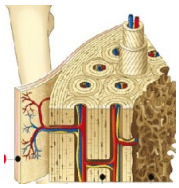
15. Diffrent types of bones in our skeleton:
    Axial skeleton consists the bones seen in the central axis of the body and the appendicular skeletons attached to it.



16. The membrane which covers the bone?
Periostium.

17. What is the function of the osteoblast cells of bones?
Deposit minerals in the bones, make them strong and firm and also help in growth and repair.

18. Structure of a typical bone:



A bone with a covering (periostium) contains blood vessels, nerves, lymph vessels, and osteoblast cells. Calcium, phosphate, collagen proteins and salts provide hardness and strength to bones. Osteoblast cells of bones deposit minerals in the bones, make them strong and firm and also help in growth and repair.

19. How is cartilage differ from bone?
The connective tissue, cartilage is softer and more flexible than bone. Due to the absence of blood vessels, its growth is slow. Nerves are also absent in cartilage.
(Cartilage at the tip of bones in the joint help to reduce friction).

20. Examples for structural frameworks in a few organisms.
Hydroskeleton, Exoskeleton & Endoskeleton.

A. **Hydroskeleton**:- Fluid filled chambers present in the body of earthworm, hydra, snail, jelly fish..
B. **Exoskeleton** :- Hard shell made of calcium carbonate in crabs, mussels and oysters.
Outer covering made of chitin in grasshoppers and cockroaches.
C. **Endoskeleton** :- Frame work of cartilages and bones in vertebrates, including man.

21. Even those with endoskeleton may also have exoskeleton parts. Examples?
Hair, fur, nails, horns, hoofs, feathers, scales.

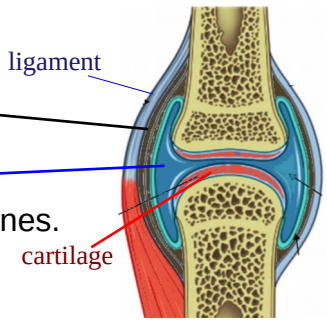
22. Examples of animals having both the endoskeleton and exoskeleton?
Turtle and crocodile. (The shell of the turtle is exoskeleton and the bones are endoskeleton.
In crocodiles, there is an exoskeleton with thick scales along with the endoskeleton)

23. The exoskeleton is sometimes shed by animals (moulting). Why?
Insects and snakes molt to form new ones as part of body growth, and others to repair damage or adapt to a changed environment.

24. Bones are connected to each other by strings like structures, called ---- ?
Ligaments.

25. The part where two or more bones are connected to each other?
Joint.

26. Structure of a typical joint:



Ligaments are found connecting two bones. Inside this the **capsule** helps in the smooth movements of bones. The synovial membrane seen inside the capsule produce the **synovial fluid** and the **cartilage** at the tip of each bone reduce friction between the bones.

27. Different types of joints and their peculiarities:

A. **Pivot joint** :- The joint at the point of skull meets the first vertebra.
This enables side to side movements of head.



B. **Ball and Socket joint**:- A joint with a socket-like and ball-shaped end seen at pectoral girdle joins with the humerus and pelvic girdle joins with the femur.
This enables movements in all planes.





C. **Hinge joint** :- A hinge-like joint in the elbow, knee, and fingers.
Enables movements towards one side like a hinge.



D. **Gliding joint** :- Joints in the wrists and ankles.
Enables gliding movements.

28. Factors required for bone growth?
Calcium – ensure hardness and strength of bones.
Vitamin D – for absorbing calcium. (mild sun exposure is good)
Protein – for bone development.

29. What is the function of bones?
Help for body movement, body shape, strength, protection and formation of blood cells.

30. What is the importance of exercise?
- Vital capacity of lugs increases, Gaseous exchange becomes efficient.
- Strengthens the limbs.
- Muscle strength increases and blood vessels increase. Bones become stronger and firm.
- Increases the efficiency of heart muscles and blood flow. Blood vessels expand.

31. Disorders of bones and muscles? Reason and symptoms?

Osteoporosis	Deficiency of Ca, vitamins and protein	Parts of bones get damaged. Bone become porous and weak.
Rheumatoid arthritis	Destruction of cartilages and synovial membrane by the immune system.	Severe pain and swelling in the joints.
Muscular dystrophy	Main reason is the change occurs in the genes.	Muscles become weak.
Sprain	Stretching or breaking of ligaments.	Pain, swelling, bruises(ചതവ്) and difficulty in moving the joints.

32. Slings and splints are used in cases of broken bones. What are these?
A sling is a system for supporting and reducing movement in case of arm and shoulder injuries.
Splint is a device to support and reduce movement when bones are fractured or sprained.
A bandage can also be used for this. (Bandage also helps to stop bleeding in the wound)

33. Bones are helpful in the study of organic evolution. Justify with an example.
Studies using DNA extracted from ancient bones (fossils) reveals the Humans' relationship with extincted hominins.

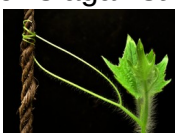
34. Different types of skeletons are arranged based on their evolutionary history :
Early hydroskeleton organisms – Exoskeleton organisms – Endoskeleton organisms
(They are more flexible for more movements and strong) - Emergence of bipedal humans.

35. Examples for stimuli that cause plant movements?
Light, water, gravity, touch, chemical substances.

36. What are the two types of movements found in plants?
Tropic movements in the direction of stimuli and **nastic movements** not in the direction of stimuli.

37. Different types of tropic movements:
a. **Phototropism** :- (eg:- Stem grows towards the direction of light and root grows against it)
b. **Geotropism** :- (eg:- Root grows towards gravity of earth and stem grows against it)
c. **Hydrotropism** :- (eg:- Root grows towards the direction of water and stem grows against it)

Other types of tropic movements:
* **Haptotropism** :- (eg:- A plant's tendrils grow toward or around the object it touches)
* **Chemotropism** :- (eg:- Pollentube moves toward the ovary where chemical is present)




38. Examples for nastic movements?
Folding up of leaves in the sensitive plant (touch-me-not) when touch on it, the leaves of some plants fold up when it gets dark, opening of flower buds, movements in predatory plants..